

Year 7 · Chapter 1 · Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

Teacher notes

Key misconception. Reading the $-$ in -5 only as "take away", so -5 feels like an action rather than a place; and thinking "opposite" means a nearby or equal number.

Big idea (fill in together). Negative numbers are to the **left** of 0 on a number line. A number and its **opposite** are the same **distance** from 0, but on the other side.

Explain 1. Why is 5 m below sea level a negative number?

Sea level is 0, and below 0 is negative.

Explain 2. How does the number line show that 6 is the opposite of -6 ?

Both are 6 steps from 0, on opposite sides of 0.

Explain 3. Why is $-(-7)$ equal to 7?

$-(-7)$ means the opposite of -7 , and the opposite of -7 is 7.

Check (hinge). What is the opposite of -8 ? Answer: **B**.

A (-8): thinks "opposite" means the same number. Re-teach with the Big idea picture: same distance, other side.

B (8): correct.

C (0): confuses opposites with a zero pair ($-8 + 8 = 0$). Point out: the opposite is a number, not the sum.

D (-7): moved one step, not to the other side. Show the two equal jumps from 0.

Next. B: page 2. A, C or D: redraw the Big idea on the board, walk out from 0 both ways, then redo Your turn 2 before page 2.

Think first. Accept " -2 ", " $B2$ " or "two down" and turn them into -2 . Point out that the lift buttons often show -1 , -2 .

Worked answers

Do now 1 Order smallest to largest: 12, 3, 30, 0
0, 3, 12, 30

Do now 2 What is $8 - 8$?
0

Do now 3 It is 0°C . It gets 4 degrees colder. What is the temperature?
 -4°C

Do now 4 What number is halfway between 0 and 10?
5

Think first A lift is at the ground floor, level 0. It goes down 2 floors to the car park. Which level is that?
level -2

Example 1 (model this) Write an integer for 5 m below sea level.
below 0: negative $\cdot = -5$

Your turn 1 Write an integer for a loss of \$8.
a loss: negative $\cdot = -8$

Example 2 (model this) The opposite of -6
 -6 is 6 steps left of 0 $\cdot$ opposite: 6

Your turn 2 The opposite of 9
9 is 9 steps right of 0 $\cdot$ opposite: -9

Example 3 (model this) $-(-7)$
= the opposite of -7 $\cdot = 7$

Your turn 3 $-(-11)$
= the opposite of -11 $\cdot = 11$

Pattern 1 opposite of 3
 -3

Pattern 2 opposite of 2
 -2

Pattern 3 opposite of 1
 -1

Pattern 4 opposite of 0
0

Pattern 5 opposite of -1
1

Pattern 6 opposite of -2
2

Pattern 7 $-(-2)$
2

Pattern 8 $-(-20)$
20

What do you notice? The number goes down by 1 each time. What happens to its opposite?
The opposite goes up by 1 each time. They move in opposite directions.

Mixed 1 What is the opposite of 7?
 -7

Mixed 2 What is the opposite of -3 ?
3

Mixed 3 What is the opposite of 12?
 -12

Mixed 4 Count back by 2s: 6, 4, 2, 0, ... What are the next two numbers?
 $-2, -4$

Mixed 5 What is the opposite of -25 ?
25

Mixed 6 $-(-4)$
4

Mixed 7 $-(-15)$
15

Mixed 8 What is the opposite of 0?
0

Mixed 9 It is 2°C . It gets 5 degrees colder. What is the temperature now?
 -3°C

Mixed 10 $-(-1)$
1

Mixed 11 Write an integer for 6 m below sea level.
 -6

Mixed 12 Write an integer for a gain of \$10.
10

Mixed 13 What number is halfway between 0 and 20?
10

Mixed 14 Which is further from 0: -7 or 5?
 -7 (7 steps, not 5)

Mixed 15 Which two integers are 10 steps from 0?
10 and -10

Go further 1 Mia says "the opposite of a number is always negative". Is she right? Explain.
No. The opposite of -3 is 3, which is positive.

Go further 2 Two opposites are 18 steps apart. What are they?
9 and -9

Go further 3 Which integers from -5 to 5 are more than 3 steps from 0?
 $-5, -4, 4, 5$

Exit 1 Write an integer for 4°C below zero.
 -4

Exit 2 What is the opposite of -13 ?
13

Exit 3 $-(-2)$
2

1.02 The number line (booklet p. 5)

Teacher notes

Key misconception. Counting marks instead of gaps (answers one too many), and subtracting the sizes ($4 - 3 = 1$) when two numbers are on opposite sides of 0.

Big idea (fill in together). Numbers get **larger** to the right. To find a distance, count the **gaps**, not the marks. If the trip passes 0, count to 0, then count on.

Explain 1. Not every mark has a number. How can you find the number at the dot?

Each gap is worth 1, so count the gaps from the nearest labelled mark.

Explain 2. Why do we add the 3 and the 5?

The trip passes 0: 3 steps to get to 0, then 5 more. Together that is 8 steps.

Explain 3. How can you check that -3 is halfway between -8 and 2 ?

-3 is 5 steps from -8 and 5 steps from 2 : the same distance from both ends.

Check (hinge). How far apart are -4 and 3 ? Answer: **B**.
A (1): did $4 - 3$, ignoring that the numbers are on opposite sides of 0. Draw it and count to 0, then on.

B (7): correct.

C (-7): right size, but a distance is never negative. Revisit the Words box.

D (-1): subtracted and kept a minus. Re-teach Example 2 on the number line.

Next. B: page 2. A or D: model counting to 0 then on, with fingers on the board line; redo Your turn 2. C: a quick class discussion on "distance is never negative".

Think first. Many will say -2 straight away. Ask how they know: "same distance from -4 and from 0 ". That is the halfway idea for Example 3.

Worked answers

Do now 1 What is the opposite of -4 ?

4

Do now 2 $-(-10)$

10

Do now 3 Write an integer for 3 floors below ground.

-3

Do now 4 Count by 5s: 15, 20, 25, ... What comes next?

30

Think first A dot is halfway between -4 and 0 . What number is it at?

-2

Example 1 (model this) Which number is at the dot?

one gap left of -2 $\therefore = -3$

Your turn 1 Which number is at the dot?

one gap right of 0 $\therefore = 1$

Example 2 (model this) How far is it from -3 to 5 ?

3 to 0, 5 from 0 $\therefore 3 + 5 = 8$

Your turn 2 How far is it from -4 to 5 ?

4 to 0, 5 from 0 $\therefore 4 + 5 = 9$

Example 3 (model this) Which integer is halfway between -8 and 2 ?

10 apart, half is 5 $\therefore 5$ right of -8 : -3

Your turn 3 Which integer is halfway between -6 and 4 ?

10 apart, half is 5 $\therefore 5$ right of -6 : -1

Pattern 1 -1 to 3

4

Pattern 2 -2 to 3

5

Pattern 3 -3 to 3

6

Pattern 4 -4 to 3

7

Pattern 5 -4 to 2

6

Pattern 6 -4 to 1

5

Pattern 7 -4 to 0

4

Pattern 8 -4 to -1

3

What do you notice? In 1 to 4, one end moves 1 step further out. In 5 to 8, it moves 1 step back in. What happens to the distance?

Moving out: the distance grows by 1 each time. Moving in: it shrinks by 1 each time.

Mixed 1 How far is it from -2 to 3 ?

5

Mixed 2 How far is it from -5 to 1 ?

6

Mixed 3 How far is it from -4 to -1 ?

3

Mixed 4 What is the opposite of -9 ?

9

Mixed 5 Start at -3 . Move 5 right. Where do you land?

2

Mixed 6 Start at 4. Move 9 left. Where do you land?

-5

Mixed 7 Start at -6 . Move 2 left. Where do you land?

-8

Mixed 8 Which integer is halfway between -4 and 2?

-1

Mixed 9 $-(-3)$

3

Mixed 10 Which integer is halfway between -9 and -1 ?

-5

Mixed 11 A number line goes $-20, -10, 0, \dots$ What are the next two numbers?

10, 20

Mixed 12 Point A is at -6 . Point B is 10 to the right of A. Where is B?

4

Mixed 13 Write an integer for a loss of \$7.

-7

Mixed 14 A lift goes from level -3 to level 6. How many floors is that?

9 floors

Mixed 15 How far is it from -8 to 7?

15

Go further 1 Two integers are 12 apart. Halfway between them is -2 . What are they?

-8 and 4

Go further 2 Start at -5 . Move 3 right, then 8 left, then 6 right. Where do you land? How many steps did you walk?

Land on -4 . $3 + 8 + 6 = 17$ steps.

Go further 3 How many integers are between -6 and 4, not counting -6 and 4?

9 (-5 to 3)

Exit 1 How far is it from -7 to 2?

9

Exit 2 Which integer is halfway between -10 and 4?

-3

Exit 3 Start at 3 and move 8 left. Where do you land?

-5

1.03 Comparing and ordering integers

(booklet p. 7)

Teacher notes

Key misconception. Magnitude confusion: " -9 is bigger than -4 because $9 > 4$ ". Students compare the digits and ignore the sign.

Big idea (fill in together). On a number line, the number further **right** is larger. So -3 is **larger** than -8 , even though 8 is more than 3. Colder temperatures are **smaller** numbers.

Explain 1. Sam says -9 is larger than -4 , because 9 is more than 4. Why is Sam wrong?

-9 is further left on the number line than -4 , so -9 is smaller.

Explain 2. How did you choose between $<$ and $>$?

The open side faces the larger number. -1 is further right, so it is larger: $-6 < -1$.

Explain 3. Why does -5 come before -2 in the list?

-5 is further left than -2 on the number line, so -5 is smaller.

Check (hinge). Which is the smallest? Answer: **C**.

A (-2): chose the "smallest digit" (magnitude confusion). Place all four on a number line together.

B (0): thinks 0 is the smallest number. Remind: negatives are less than 0.

C (-9): correct.

D (-5): partly there; check each pair on a number line.

Next. C: page 2. A or D: put all four numbers on one number line on the board, and read left to right. B: short reminder that negatives are below 0 (thermometer).

Think first. Nearly all will say -8°C . Ask: "So which number is smaller?" This is the Big idea: colder = lower = smaller.

Worked answers

Do now 1 How far is it from -5 to 3 ?

8

Do now 2 Start at -1 and move 6 left. Where do you land?

-7

Do now 3 What is the opposite of -15 ?

15

Do now 4 Which integer is halfway between -6 and 2 ?

-2

Think first Which is colder: -8°C or -3°C ?

-8°C

Example 1 (model this) Which is larger: -9 or -4 ?

-4 is right of -9 · larger: -4

Your turn 1 Which is larger: -2 or -7 ?

-2 is right of -7 · larger: -2

Example 2 (model this) Write $<$ or $>$: -6 $\square$ -1

-6 is left of -1 · $-6 < -1$

Your turn 2 Write $<$ or $>$: 0 $\square$ -5

0 is right of -5 · $0 > -5$

Example 3 (model this) Order smallest to largest: $3, -5, 0, -2$

smallest: -5 · $-5, -2, 0, 3$

Your turn 3 Order smallest to largest: $-1, 4, -6, 2$

smallest: -6 · $-6, -1, 2, 4$

Pattern 1 3 $\square$ 5

$<$

Pattern 2 -3 $\square$ 5

$<$

Pattern 3 -3 $\square$ -5

$>$

Pattern 4 3 $\square$ -5

$>$

Pattern 5 -5 $\square$ -3

$<$

Pattern 6 -30 $\square$ -5

$<$

Pattern 7 -5 $\square$ 0

$<$

Pattern 8 0 $\square$ -30

$>$

What do you notice? Look at 1 and 3. Both numbers became negative. What happened to the symbol?

It flipped: $3 < 5$, but $-3 > -5$.

Mixed 1 Write $<$ or $>$: -4 $\square$ 2

$<$

Mixed 2 Write $<$ or $>$: -6 $\square$ -1

$<$

Mixed 3 Write $<$ or $>$: 0 $\square$ -3

$>$

Mixed 4 How far is it from -11 to -2 ?

9

Mixed 5 Write $<$ or $>$: -8 $\square$ -9

$>$

Mixed 6 Write $<$ or $>$: -20 $\square$ -19

$<$

Mixed 7 Which is smaller: -7 or -3 ?

-7

Mixed 8 Which is larger: -15 or -51 ?

-15

Mixed 9 What is the opposite of -14 ?

14

Mixed 10 Order smallest to largest: $3, -2, -5$

$-5, -2, 3$

Mixed 11 Order smallest to largest: $-1, -8, 0, -4$

$-8, -4, -1, 0$

Mixed 12 Which is colder: -15°C or -12°C ?

-15°C

Mixed 13 Which integer is halfway between -10 and 2 ?

-4

Mixed 14 Write an integer that is between -5 and -3 .

-4

Mixed 15 Order largest to smallest: $-4, 7, -10, 0$

$7, 0, -4, -10$

Go further 1 Always, sometimes or never? "A negative number is smaller than a positive number." Explain.

Always. Every negative is left of 0 and every positive is right of 0.

Go further 2 I am an odd integer. I am less than -2 and greater than -6 . What could I be?

-3 or -5

Go further 3 Order these coldest first: $-7^{\circ}\text{C}, 4^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}$. How far apart are the coldest and warmest?

$-12, -7, 0, 4$. 16 degrees apart.

Exit 1 Write $<$ or $>$: -8 $\square$ -3

$<$

Exit 2 Order smallest to largest: $2, -9, -1, -4$

$-9, -4, -1, 2$

Exit 3 Which is warmer: -6°C or -10°C ?

-6°C

1.04 Adding integers (booklet p. 9)

Teacher notes

Key misconception. Treating $-2 + 5$ as " $-(2 + 5)$ " (the symmetric error), and carrying "two negatives make a positive" over from multiplication into addition.

Big idea (fill in together). Start at the **first** number. Adding a positive: move **right**. Adding a negative: move **left**, which is the same as **subtracting**.

Explain 1. Why do we move right for $-5 + 3$?

We are adding a positive number (3), and adding a positive moves right.

Explain 2. Why do $2 + (-5)$ and $2 - 5$ give the same answer?

Both start at 2 and move 5 left, to -3. Adding a negative is the same as subtracting.

Explain 3. Tom says $-4 + (-3) = 7$. What has Tom done wrong?

Tom used "two negatives make a positive", which is a rule for multiplying. Adding -3 moves left: $-4 + (-3) = -7$.

Check (hinge). Work out $-2 + 5$ Answer: B .

A (-7): added the sizes and kept the minus sign ("symmetric" error). Re-teach with Example 1 on the number line.

B (3): correct.

C (-3): moved the wrong way, or did $5 - 2$ and kept the minus. Act it out: start at -2 , walk 5 steps right.

D (7): ignored the negative sign. Check they read -2 as a place left of zero (Words box).

Next. Most students B: go on to page 2. Students with A, C or D: a short re-teach at the board with Examples 1 and 2 on the number line (walk it out), then Your turn 1 and 2 again before page 2. Students who finish Mixed practice: Go further.

Think first. Take answers without judging. Show a thermometer: start at -3 , count up 5. Link to Example 1: the same move on a sideways number line.

Worked answers

Do now 1 $-7 \square -2$ Write $<$ or $>$.

$-7 < -2$

Do now 2 Order smallest to largest: 4, -6 , 0, -1

$-6, -1, 0, 4$

Do now 3 What is the opposite of -9 ?

9

Do now 4 Start at -4 . Move 6 right. Where do you land?

2

Think first It is -3°C at 6 am. By noon it is 5°C warmer. What is the temperature at noon?

2°C

Example 1 (model this) $-5 + 3$

start at -5 , move 3 right $\cdot = -2$

Your turn 1 $-6 + 4$

start at -6 , move 4 right $\cdot = -2$

Example 2 (model this) $2 + (-5)$

start at 2, move 5 left $\cdot = -3$

Your turn 2 $3 + (-7)$

start at 3, move 7 left $\cdot = -4$

Example 3 (model this) $-4 + (-3)$

$= -4 - 3 \cdot = -7$

Your turn 3 $-2 + (-6)$

$= -2 - 6 \cdot = -8$

Pattern 1 $-3 + 3$

0

Pattern 2 $-3 + 2$

-1

Pattern 3 $-3 + 1$

-2

Pattern 4 $-3 + 0$

-3

Pattern 5 $-3 + (-1)$

-4

Pattern 6 $-3 + (-2)$

-5

Pattern 7 $-3 + (-3)$

-6

Pattern 8 $-30 + (-30)$

-60

What do you notice? The number being added goes down by 1 each time. What happens to the answer?

The answer goes down by 1 too. Adding a negative carries on the pattern: it makes the answer smaller.

Mixed 1 $-5 + 2$

-3

Mixed 2 $-3 + 7$

4

Mixed 3 $-8 + 8$

0

Mixed 4 Write < or >: $-8 \square -5$

<

Mixed 5 $4 + (-6)$

-2

Mixed 6 $2 + (-9)$

-7

Mixed 7 $-3 + (-4)$

-7

Mixed 8 $-7 + (-2)$

-9

Mixed 9 How far apart are -8 and 5 ?

13

Mixed 10 $15 + (-20)$

-5

Mixed 11 $-12 + 5$

-7

Mixed 12 A diver is at -12 m. She rises 5 m. Where is she now?

$-12 + 5 = -7$ m

Mixed 13 What is the opposite of -12 ?

12

Mixed 14 $-6 + \square = -1$

$\square = 5$

Mixed 15 It is -9°C . It rises 4°C , then 7°C more. What is the temperature now?

$-9 + 4 + 7 = 2^\circ\text{C}$

Go further 1 Always, sometimes or never true? "Adding a negative number makes the answer smaller." Explain.

Always. Adding a negative always moves left, so the answer is always smaller than the starting number.

Go further 2 Find two integers that add to -3 . Then find two more, where one of them is positive.

e.g. $-1 + (-2) = -3$; e.g. $2 + (-5) = -3$

Go further 3 Start at -4 . Add 2 , 6 and -5 in any order. Where do you finish? Does the order matter?

$-4 + 2 + 6 + (-5) = -1$. The order does not matter.

Exit 1 $-7 + 3$

-4

Exit 2 $4 + (-9)$

-5

Exit 3 It is -6°C . It warms up 10°C . What is the temperature now?

$-6 + 10 = 4^\circ\text{C}$

1.05 Subtracting integers (booklet p. 11)

Teacher notes

Key misconception. Treating $-(-3)$ as -3 , so $4 - (-3) = 1$. Students see one minus sign and ignore the other.

Big idea (fill in together). Subtracting is the same as adding the **opposite**. $5 - 3 = 5 + (-3)$: move **left**. $5 - (-3) = 5 + 3$: move **right**.

Explain 1. Why is the answer to $2 - 6$ negative?

We take away more (6) than we start with (2), so we go past 0.

Explain 2. Why can we rewrite $3 - (-4)$ as $3 + 4$?

Subtracting is adding the opposite, and the opposite of -4 is 4 .

Explain 3. Ali says $3 - (-2) = 1$. What has Ali done wrong?

Ali took away 2, not -2 . Subtracting -2 means adding 2: $3 - (-2) = 3 + 2 = 5$.

Check (hinge). Work out $4 - (-3)$ Answer: B.

A (1): treated $-(-3)$ as -3 (ignored the second negative). Re-teach Example 2: rewrite as $+3$.

B (7): correct.

C (-7): added the sizes and made it negative. Walk it on the number line.

D (-1): did $3 - 4$. Check they start at the first number.

Next. B: page 2. A: rewrite on the board, "subtract negative three = add three", and do Pattern 1-7 together. C or D: walk it on a floor or board number line.

Think first. Link colder to subtracting: $3 - 5 = -2$. Then ask: what would "5 degrees less cold" mean? (Leads to subtracting a negative.)

Worked answers

Do now 1 $-7 + 4$
 -3

Do now 2 $5 + (-8)$
 -3

Do now 3 Write $<$ or $>$: $-6 \square -2$
 $<$

Do now 4 What is the opposite of -5 ?
 5

Think first It is 3°C . It gets 5 degrees colder. What is the temperature now?
 -2°C

Example 1 (model this) $2 - 6$
start at 2, move 6 left $\cdot = -4$

Your turn 1 $-1 - 4$
start at -1 , move 4 left $\cdot = -5$

Example 2 (model this) $3 - (-4)$
 $= 3 + 4 \cdot = 7$

Your turn 2 $-5 - (-2)$
 $= -5 + 2 \cdot = -3$

Example 3 (model this) $-6 - (-9)$
 $= -6 + 9 \cdot = 3$

Your turn 3 $-8 - (-3)$
 $= -8 + 3 \cdot = -5$

Pattern 1 $5 - 3$
 2

Pattern 2 $5 - 2$
 3

Pattern 3 $5 - 1$
 4

Pattern 4 $5 - 0$
 5

Pattern 5 $5 - (-1)$
 6

Pattern 6 $5 - (-2)$
 7

Pattern 7 $5 - (-3)$
 8

Pattern 8 $-5 - (-3)$
 -2

What do you notice? The number being subtracted goes down by 1 each time. What happens to the answer?
The answer goes up by 1 each time. So subtracting a negative makes the answer larger.

Mixed 1 $3 - 7$
 -4

Mixed 2 $0 - 5$
 -5

Mixed 3 $-2 - 4$
 -6

Mixed 4 $-3 + (-6)$
 -9

Mixed 5 $5 - (-2)$
 7

Mixed 6 $-3 - (-8)$
 5

Mixed 7 $-4 - (-4)$
 0

Mixed 8 $-9 - (-2)$
 -7

Mixed 9 Write $<$ or $>$: $-12 \square -8$
 $<$

Mixed 10 $-12 - 8$
 -20

Mixed 11 $7 - (-7)$
 14

Mixed 12 It is -4°C . It gets 7°C colder. What is the temperature now?
 $-4 - 7 = -11^{\circ}\text{C}$

Mixed 13 $-6 + 10$
 4

Mixed 14 How much warmer is 6°C than -2°C ?
 $6 - (-2) = 8^{\circ}\text{C}$

Mixed 15 $10 - (-5) - 20$
 $= 15 - 20 = -5$

Go further 1 Is $7 - (-2)$ the same as $7 + 2$? Use a number line to explain.
Yes. Both start at 7 and move 2 right, to 9.

Go further 2 Find a positive integer and a second integer whose difference (first $-$ second) is -5 .
e.g. $1 - 6 = -5$

Go further 3 A submarine is at -120 m. A plane is at 350 m. How far apart are they?
 $350 - (-120) = 470$ m

Exit 1 $-2 - 7$
 -9

Exit 2 $6 - (-5)$
 11

Exit 3 $-4 - (-9)$
 5

1.06 Multiplying integers (booklet p. 13)

Teacher notes

Key misconception. Believing a negative $\times$ a negative is negative ("negative means less"), and, from the other side, using "two negatives make a positive" when adding.

Big idea (fill in together). Multiply the numbers, then choose the sign. **Same signs:** the answer is **positive**. **Different signs:** the answer is **negative**.

Explain 1. Why is $3 \times (-2)$ negative?

$3 \times (-2)$ means 3 groups of -2 . Adding negatives gives a negative: -6 .

Explain 2. Why does -5×4 give the same answer as $4 \times (-5)$?

Swapping the order of a multiplication does not change the answer. Both are 4 groups of -5 , which is -20 .

Explain 3. $1 \times (-6) = -6$ and $0 \times (-6) = 0$: the answer went up by 6. What will $-1 \times (-6)$ be?

6. Each time the first number goes down by 1, the answer goes up by 6. So a negative $\times$ a negative is positive.

Check (hinge). Work out $-4 \times (-5)$ Answer: **B**.

A (-20): thinks negatives always give a negative. Go through Spot the pattern now, together.

B (20): correct.

C (-9): added instead of multiplying ($-4 + (-5)$). Check the operation.

D (9): added and then used the sign rule. Check the operation.

Next. B: page 2. A: do Spot the pattern together on the board before page 2. C or D: underline the operation sign in each question before answering.

Think first. Write $4 \times (-3) = -12$ from their answers. This is Example 1: repeated addition of a negative.

Worked answers

Do now 1 $-3 - 5$

-8

Do now 2 $2 - (-6)$

8

Do now 3 $(-4) + (-4) + (-4)$

-12

Do now 4 7×8

56

Think first You lose \$3 a day for 4 days. By how much has your money changed?

$-\$12$ ($4 \times (-3)$)

Example 1 (model this) $3 \times (-2)$

$(-2) + (-2) + (-2) = -6$

Your turn 1 $4 \times (-3)$

$(-3) + (-3) + (-3) + (-3) = -12$

Example 2 (model this) -5×4

different signs: $- \cdot 5 \times 4 = 20$, so -20

Your turn 2 $6 \times (-7)$

different signs: $- \cdot 6 \times 7 = 42$, so -42

Example 3 (model this) $-3 \times (-6)$

same signs: $+ \cdot 3 \times 6 = 18$, so 18

Your turn 3 $-8 \times (-5)$

same signs: $+ \cdot 8 \times 5 = 40$, so 40

Pattern 1 $3 \times (-4)$

-12

Pattern 2 $2 \times (-4)$

-8

Pattern 3 $1 \times (-4)$

-4

Pattern 4 $0 \times (-4)$

0

Pattern 5 $-1 \times (-4)$

4

Pattern 6 $-2 \times (-4)$

8

Pattern 7 $-3 \times (-4)$

12

Pattern 8 -3×4

-12

What do you notice? In 1 to 7, the answer goes up by the same amount each time. What does that show about negative $\times$ negative?

It goes up by 4 each time, so $-1 \times (-4) = 4$. Negative $\times$ negative is positive.

Mixed 1 $4 \times (-3)$

-12

Mixed 2 -5×2

-10

Mixed 3 $-6 \times (-2)$

12

Mixed 4 $-7 + (-3)$

-10

Mixed 5 $7 \times (-4)$

-28

Mixed 6 $-3 \times (-9)$

27

Mixed 7 -8×5

-40

Mixed 8 $-7 \times (-3)$

21

Mixed 9 $-6 - (-2)$

-4

Mixed 10 $-10 \times (-4)$

40

Mixed 11 $-2 \times 3 \times (-4)$

$= -6 \times (-4) = 24$

Mixed 12 A diver goes down 5 m every minute for 6 minutes. What is her change in height?

$6 \times (-5) = -30$ m

Mixed 13 $(-3) + (-3) + (-3) + (-3)$

-12

Mixed 14 $\square \times (-4) = 20$

$\square = -5$

Mixed 15 Which is larger: -2×6 or $-2 \times (-6)$?

$-2 \times (-6)$, which is 12 (-2×6 is -12)

Go further 1 Find all the pairs of integers that multiply to -12.

1 and -12, -1 and 12, 2 and -6, -2 and 6, 3 and -4, -3 and 4

Go further 2 Always, sometimes or never? "Negative $\times$ negative $\times$ negative is negative." Explain.

Always. The first two give a positive; positive $\times$ negative is negative.

Go further 3 Work out $(-1) \times (-1) \times (-1) \times (-1) \times (-1)$. What if there were 100 of them?

-1. With 100 of them, +1 (they pair up).

Exit 1 -6×4

-24

Exit 2 $-9 \times (-2)$

18

Exit 3 The temperature falls 3°C each hour for 5 hours. What is the total change?

$5 \times (-3) = -15^\circ\text{C}$

1.07 Dividing integers (booklet p. 15)

Teacher notes

Key misconception. Keeping the negative when both numbers are negative ($-20 \div (-4) = -5$), usually from not linking $\div$ to $\times$ facts.

Big idea (fill in together). Division **undoes** multiplication, so the sign rules are the **same**. Same signs: **positive**. Different signs: **negative**.

Explain 1. How does knowing $3 \times (-4) = -12$ help you work out $-12 \div 3$?

Division undoes multiplication: 3 groups of -4 make -12, so -12 shared into 3 groups is -4.

Explain 2. How can you check that $28 \div (-4) = -7$?

Multiply back: $-7 \times (-4) = 28$, so the answer is right.

Explain 3. Jo says $-30 \div (-6) = -5$. What has Jo done wrong?

The signs are the same, so the answer is positive: 5. Check: $5 \times (-6) = -30$.

Check (hinge). Work out $-20 \div (-4)$ Answer: **B**.

A (-5): kept the negative. Check by multiplying back: $-5 \times (-4) = 20$, not -20.

B (5): correct.

C (-16): subtracted: $-20 - (-4)$. Check the operation.

D (80): multiplied. Check the operation.

Next. B: page 2. A: model "multiply back to check" on three examples. C or D: underline the operation before answering.

Think first. Write $-24 \div 4 = -6$: the share of a debt is itself a debt. Link to Example 1.

Worked answers

Do now 1 -6×7
-42

Do now 2 $-4 \times (-8)$
32

Do now 3 $-2 - 9$
-11

Do now 4 Order smallest to largest: $-4, 2, -8$
-8, -4, 2

Think first A debt of \$24 is shared equally by 4 people. How much does each person owe?
\$6 each ($-24 \div 4 = -6$)

Example 1 (model this) $-12 \div 3$
 $3 \times (-4) = -12$ · so $-12 \div 3 = -4$

Your turn 1 $-15 \div 5$
 $5 \times (-3) = -15$ · so $-15 \div 5 = -3$

Example 2 (model this) $28 \div (-4)$
different signs: $-$ · $28 \div 4 = 7$, so -7

Your turn 2 $-36 \div 9$
different signs: $-$ · $36 \div 9 = 4$, so -4

Example 3 (model this) $-30 \div (-6)$
same signs: $+$ · $30 \div 6 = 5$, so 5

Your turn 3 $-48 \div (-8)$
same signs: $+$ · $48 \div 8 = 6$, so 6

Pattern 1 $12 \div 3$
4

Pattern 2 $-12 \div 3$
-4

Pattern 3 $12 \div (-3)$
-4

Pattern 4 $-12 \div (-3)$
4

Pattern 5 $-12 \div (-4)$
3

Pattern 6 $-12 \div (-6)$
2

Pattern 7 $-12 \div 6$
-2

Pattern 8 $-120 \div (-6)$
20

What do you notice? Which answers are negative? What do their questions have in common?
2, 3 and 7. Each one has different signs.

Mixed 1 $-12 \div 3$
-4

Mixed 2 $18 \div (-6)$
-3

Mixed 3 $-24 \div (-8)$
3

Mixed 4 -9×3
-27

Mixed 5 $-35 \div 7$
-5

Mixed 6 $42 \div (-7)$
-6

Mixed 7 $-36 \div (-4)$
9

Mixed 8 $-64 \div (-8)$
8

Mixed 9 $-4 - (-10)$
6

Mixed 10 $81 \div (-9)$
-9

Mixed 11 $-100 \div (-25)$
4

Mixed 12 $\square \div 4 = -5$
 $\square = -20$

Mixed 13 $-5 \times (-7)$
35

Mixed 14 The temperature fell 18°C in 6 hours, the same each hour. What was the change each hour?
 $-18 \div 6 = -3^\circ\text{C}$

Mixed 15 $-80 \div 4 \div (-2)$
 $= -20 \div (-2) = 10$

Go further 1 Find the average of $-6, 2, -5$ and -3 . (Add them, then divide by 4.)
Total -12 , so $-12 \div 4 = -3$

Go further 2 Write three different divisions that have the answer -4 .
e.g. $-8 \div 2, 12 \div (-3), -20 \div 5$

Go further 3 Always, sometimes or never? "Dividing makes a number smaller." Explain.
Sometimes. $-12 \div 3 = -4$, and -4 is larger than -12 .

Exit 1 $-35 \div 5$
-7

Exit 2 $-72 \div (-8)$
9

Exit 3 A debt of \$60 is shared by 3 people. Write each share as an integer.
 $-60 \div 3 = -20$

1.08 Order of operations with integers

(booklet p. 17)

Teacher notes

Key misconception. Working strictly left to right ($10 - 3 \times (-2) = -14$), and losing a negative when subtracting a negative product.

Big idea (fill in together). First, work out the **brackets**. Next, $\times$ and $\div$, from **left to right**. Last, $+$ and $-$, from left to right. Use the integer rules at every **step**.

Explain 1. Why do we work out 4×2 before the adding?
In the order of operations, $\times$ and $\div$ come before $+$ and $-$.

Explain 2. Without brackets, $-3 + 7 \times 2 = 11$. Why do the brackets change the answer?

Brackets are worked out first, so we add before multiplying: $4 \times 2 = 8$.

Explain 3. Mia says $6 - 2 \times (-3) = -12$. What has Mia done wrong?

Mia worked out $6 - 2$ first. $\times$ comes first: $6 - (-6) = 6 + 6 = 12$.

Check (hinge). Work out $10 - 3 \times (-2)$ Answer: **C**.

A (-14): worked left to right: $10 - 3 = 7$, $7 \times (-2) = -14$. Re-teach the order.

B (4): did $\times$ first but lost the second negative: $10 - 6$. Re-teach Example 3.

C (16): correct.

D (-16): sign errors at both steps. Do it step by step on the board.

Next. C: page 2. A: re-teach the order with the Think first example. B or D: model Example 3, one line at a time, saying each step aloud.

Think first. Both methods are "reasonable"; the order of operations is an agreement so everyone gets the same answer. 14 is right.

Worked answers

Do now 1 $-18 \div 3$

-6

Do now 2 $-4 \times (-6)$

24

Do now 3 $5 - 9$

-4

Do now 4 $-8 + 3$

-5

Think first Two students work out $2 + 3 \times 4$. One gets 20. One gets 14. Who is right?

14 ($\times$ first: $2 + 12$)

Example 1 (model this) $-3 + 4 \times 2$

$= -3 + 8 = 5$

Your turn 1 $-10 + 3 \times 2$

$= -10 + 6 = -4$

Example 2 (model this) $(-3 + 7) \times 2$

$= 4 \times 2 = 8$

Your turn 2 $(1 - 6) \times 3$

$= -5 \times 3 = -15$

Example 3 (model this) $6 - 2 \times (-3)$

$= 6 - (-6) = 6 + 6 = 12$

Your turn 3 $-5 - 4 \times (-2)$

$= -5 - (-8) = -5 + 8 = 3$

Pattern 1 $2 + 3 \times 4$

14

Pattern 2 $2 - 3 \times 4$

-10

Pattern 3 $-2 - 3 \times 4$

-14

Pattern 4 $-2 - 3 \times (-4)$

10

Pattern 5 $(-2 - 3) \times (-4)$

20

Pattern 6 $(-2 + 3) \times (-4)$

-4

Pattern 7 $-2 + 3 \times (-4)$

-14

Pattern 8 $-20 + 30 \times (-4)$

-140

What do you notice? Find two questions with the same numbers but different answers. What made them different?
e.g. 4 and 5: $-2 - 3 \times (-4) = 10$ but $(-2 - 3) \times (-4) = 20$. The brackets changed the order.

Mixed 1 $-4 + 2 \times 3$

$= -4 + 6 = 2$

Mixed 2 $5 - 3 \times 4$

$= 5 - 12 = -7$

Mixed 3 $-2 \times 3 + 1$

$= -6 + 1 = -5$

Mixed 4 $-12 \div (-4)$

3

Mixed 5 $(-2 + 5) \times 2$

$= 3 \times 2 = 6$

Mixed 6 $(1 - 4) \times 3$

$= -3 \times 3 = -9$

Mixed 7 $-12 \div 3 + 6$

$= -4 + 6 = 2$

Mixed 8 $8 - 10 \div 2$

$= 8 - 5 = 3$

Mixed 9 $-6 \times (-5)$

30

Mixed 10 $-3 \times (-2) - 7$

$= 6 - 7 = -1$

Mixed 11 $(-6 - 2) \div 4$

$= -8 \div 4 = -2$

Mixed 12 $6 - 2 \times (-5)$

$= 6 - (-10) = 16$

Mixed 13 $-3 - (-9)$

6

Mixed 14 $-2 \times 3 + 4 \times (-2)$

$= -6 + (-8) = -14$

Mixed 15 $-3 \times (2 - 6)$

$= -3 \times (-4) = 12$

Go further 1 Put brackets in to make it true: $-4 + 6 \times 3 = 6$

$(-4 + 6) \times 3 = 6$

Go further 2 Use -2 , 3 and 5 once each, with $+$ and $\times$, and brackets if you like. What is the smallest answer you can make?

$-2 \times (3 + 5) = -16$

Go further 3 Write a calculation with brackets, $\times$ and $-$ that has the answer -12 .

e.g. $(2 - 5) \times 4 = -12$

Exit 1 $-6 + 2 \times 4$

$= -6 + 8 = 2$

Exit 2 $(-6 + 2) \times 4$

$= -4 \times 4 = -16$

Exit 3 $9 - 3 \times (-2)$

$= 9 - (-6) = 15$

1.09 Integers on a calculator (booklet p. 19)

Teacher notes

Key misconception. Using the subtract key for a negative number, and trusting the calculator display without checking the sign or size.

Big idea (fill in together). Use the $(-)$ key to make a number negative, and the $-$ key to subtract. **Estimate** first. Then check the calculator's answer has the right **sign** and size.

Explain 1. Why do we press $(-)$ at the start, and not the $-$ key?

The $(-)$ key makes 37 negative. The $-$ key subtracts, and on many calculators it would take 37 away from the last answer.

Explain 2. How does the estimate show that $-1\ 008$ is a sensible answer?

The estimate is $-1\ 000$: the same sign as $-1\ 008$ and close in size.

Explain 3. Why must you type the brackets?

Without brackets, the calculator follows the order of operations and works out 47×12 first.

Check (hinge). Ella presses $(-) 6 - 4 =$ What does her calculator show? Answer: **C**.

A (-2) : read it as $-6 + 4$ or as $6 - 4$ with a minus. Say it aloud: "negative six, subtract four".

B (2) : ignored the $(-)$ key. Re-teach the two keys.

C (-10) : correct.

D (10) : ignored the negative and added. Re-teach the two keys.

Next. C: page 2. A, B or D: everyone presses $(-) 6 - 4 =$ together and says it aloud. Check which key each calculator model uses (some use $+/-$ after the number).

Think first. Round to -50×20 . The point: an estimate tells you the size and the sign before you touch the calculator.

Worked answers

Do now 1 $-3 + 2 \times 5$

7

Do now 2 $(4 - 9) \times 2$

-10

Do now 3 $-56 \div 8$

-7

Do now 4 $-7 \times (-6)$

42

Think first Without a calculator: is -48×21 about -100, about -1 000, or about -10 000?

About -1 000 (-50×20)

Example 1 (model this) $-37 + 85$

keys: $(-) 37 + 85 = \cdot = 48$

Your turn 1 $-126 + 59$

keys: $(-) 126 + 59 = \cdot = -67$

Example 2 (model this) -48×21

estimate: $-50 \times 20 = -1\ 000 \quad \cdot = -1\ 008$

Your turn 2 -39×31

estimate: $-40 \times 30 = -1\ 200 \quad \cdot = -1\ 209$

Example 3 (model this) $(-35 - 47) \times 12$

bracket: $-82 \quad \cdot = -984$

Your turn 3 $(-28 + 13) \times (-6)$

bracket: $-15 \quad \cdot = 90$

Pattern 1 -25×4

-100

Pattern 2 -25×8

-200

Pattern 3 -25×12

-300

Pattern 4 -25×16

-400

Pattern 5 $-25 \times (-16)$

400

Pattern 6 $25 \times (-16)$

-400

Pattern 7 -250×16

-4 000

Pattern 8 $-250 \times (-160)$

40 000

What do you notice? In 1 to 4, the second number goes up by 4 each time. What happens to the answer?

It goes down by 100 each time: one more group of $4 \times (-25) = -100$.

Mixed 1 $-347 + 128$

-219

Mixed 2 $-95 - 187$

-282

Mixed 3 -56×23

-1 288

Mixed 4 No calculator: $-6 + 3 \times (-2)$

$= -6 + (-6) = -12$

Mixed 5 $68 \times (-45)$

-3 060

Mixed 6 $-37 \times (-29)$

1 073

Mixed 7 $-2\ 016 \div 42$

-48

Mixed 8 $-1\ 248 \div (-24)$

52

Mixed 9 No calculator: $-45 \div (-9)$

5

Mixed 10 $(-64 - 36) \times 15$

$= -100 \times 15 = -1\ 500$

Mixed 11 Estimate -62×19 . Then use your calculator.

$\approx -60 \times 20 = -1\ 200$; -1 178

Mixed 12 A submarine at -185 m rises 12 m a minute for 9 minutes. Where is it now?

$-185 + 9 \times 12 = -77$ m

Mixed 13 No calculator: $-8 \times (-7)$

56

Mixed 14 Find the missing number: $-23 \times \square = 1\ 403$

$\square = 1\ 403 \div (-23) = -61$

Mixed 15 $(-18 + 7) \times (-33)$

$= -11 \times (-33) = 363$

Go further 1 Use your calculator to find the missing number:

$-37 \times \square = 1\ 147$

$\square = 1\ 147 \div (-37) = -31$

Go further 2 A student gets 1 008 for -48×21 . How can you tell it is wrong without working it out?

Different signs, so the answer must be negative. It should be -1 008.

Go further 3 Moscow had these temperatures: -12, -8, -15, -3 and -7°C . Find the average.

Total -45, so $-45 \div 5 = -9^\circ\text{C}$

Exit 1 $-234 + 97$

-137

Exit 2 $-38 \times (-27)$

1 026

Exit 3 $(-45 - 27) \div 8$

$= -72 \div 8 = -9$

1.10 Integer problems (booklet p. 21)

Worked answers

Teacher notes

Key misconception. Choosing the operation from key words alone, and treating “how far apart” as a subtraction of sizes when the two numbers are on opposite sides of 0.

Big idea (fill in together). Read the story, then write a number sentence. Up, rise, earn: + . Down, fall, spend: – . The same change again and again: $\times$.

Explain 1. How did you know whether to add or subtract?
Warmer means the temperature goes up, so add. Colder means it goes down, so subtract.

Explain 2. What does a balance of $-\$25$ mean for Sam?
Sam owes the bank \$25.

Explain 3. Why do we multiply before we add?
The diver goes down 3 m four times, so the total change is $4 \times (-3) = -12$. Then add the change to the start.

Check (hinge). A gull is at 15 m. A shark is at -20 m. How far apart are they? Answer: **B** .

A (5 m): did $20 - 15$, ignoring that they are on opposite sides of 0. Draw a vertical number line.

B (35 m): correct: $15 - (-20) = 35$.

C (-5 m): did $15 + (-20)$. A distance is higher – lower, and never negative.

D (-35 m): right size, but a distance is never negative.

Next. B: page 2. A, C or D: draw a vertical number line (sea level at 0) and count 15 up and 20 down.

Think first. $-2 + 6 - 3 = 1$. Ask students to act it out with a finger on a vertical number line.

Do now 1 $-144 \div 12$

-12

Do now 2 $-2 \times (3 - 7)$

8

Do now 3 $-5 - (-8)$

3

Do now 4 Which is warmer: -14°C or -9°C ?

-9°C

Think first A lift at level -2 goes up 6 floors, then down 3 floors. Which level is it on now?

level 1

Example 1 (model this) It is -4°C at 6 am. By noon it is 9°C warmer. What is the temperature at noon?

$-4 + 9 = 5^{\circ}\text{C}$

Your turn 1 It is 3°C at 6 pm. By midnight it is 8°C colder. What is the temperature at midnight?

$3 - 8 = -5^{\circ}\text{C}$

Example 2 (model this) Sam has \$45. Sam spends \$70. What is the balance?

$45 - 70 = -\$25$

Your turn 2 Ana has $-\$30$. She is paid \$55. What is her balance?

$-30 + 55 = \$25$

Example 3 (model this) A diver at -10 m goes down 3 m a minute for 4 minutes. Where is she now?

$4 \times (-3) = -12$ $-10 + (-12) = -22\text{ m}$

Your turn 3 A diver at -40 m rises 5 m a minute for 6 minutes. Where is she now?

$6 \times 5 = 30$ $-40 + 30 = -10\text{ m}$

Pattern 1 $-5 + 8$

3

Pattern 2 $-5 + 5$

0

Pattern 3 $-5 + 2$

-3

Pattern 4 $-5 - 2$

-7

Pattern 5 $-5 - (-2)$

-3

Pattern 6 -5×2

-10

Pattern 7 $-5 \times (-2)$

10

Pattern 8 $-5 \div (-5)$

1

What do you notice? Which two questions have the same answer? Why?

3 and 5 are both -3 : subtracting -2 is the same as adding 2.

Mixed 1 It is -5°C . It rises 8°C . What is the temperature now?

$-5 + 8 = 3^{\circ}\text{C}$

Mixed 2 It is 4°C . It falls 9°C . What is the temperature now?

$4 - 9 = -5^{\circ}\text{C}$

Mixed 3 A lift is at level -3 . It goes up 5 floors. Which level is it on?

$-3 + 5 = \text{level } 2$

Mixed 4 Order coldest first: -3°C , -13°C , 3°C , 0°C

$-13, -3, 0, 3$

Mixed 5 You have \$10. You spend \$25. What is your balance?

$10 - 25 = -\$15$

Mixed 6 Your balance is $-\$20$. You earn \$30. What is it now?

$-20 + 30 = \$10$

Mixed 7 How far apart are 15 m and -10 m ?

$15 - (-10) = 25\text{ m}$

Mixed 8 How far apart are -4 m and -19 m ?

$-4 - (-19) = 15\text{ m}$

Mixed 9 Which integer is halfway between -12 and 4 ?

-4

Mixed 10 A diver goes down 3 m a minute for 5 minutes. What is the change in depth?

$5 \times (-3) = -15\text{ m}$

Mixed 11 It falls 2°C an hour for 6 hours. What is the total change?

$6 \times (-2) = -12^{\circ}\text{C}$

Mixed 12 A balance is \$20. You spend \$35, then earn \$10. What is the balance?

$20 - 35 + 10 = -\$5$

Mixed 13 $-4 + 3 \times (-5)$

$= -4 + (-15) = -19$

Mixed 14 Quiz: $+3$ if right, -1 if wrong. Kai gets 6 right and 5 wrong. His score?

$6 \times 3 + 5 \times (-1) = 18 + (-5) = 13$

Mixed 15 A plane is at 900 m . A submarine is at -150 m . How far apart are they?

$900 - (-150) = 1\,050\text{ m}$

Go further 1 In the quiz, Lea answered 10 questions and scored 14. How many were right?

6 right: $6 \times 3 + 4 \times (-1) = 18 - 4 = 14$

Go further 2 It is -6°C . It warms 2°C an hour for 12 hours, then cools 3°C an hour for 4 hours. Temperature?

$-6 + 12 \times 2 - 4 \times 3 = -6 + 24 - 12 = 6^{\circ}\text{C}$

Exit 1 It is -7°C . It rises 11°C . What is the temperature now?

$-7 + 11 = 4^{\circ}\text{C}$

Exit 2 A balance is \$15. You spend \$40. What is the balance?

$15 - 40 = -\$25$

Exit 3 A diver at -8 m goes down 2 m a minute for 5 minutes. Where is she?

$-8 + 5 \times (-2) = -18\text{ m}$

Review Chapter review (booklet p. 23)

Worked answers

1 What is the opposite of -17 ?

17

2 $-(-8)$

8

3 How far is it from -6 to 9 ?

15

4 Which integer is halfway between -14 and 2 ?

-6

5 Write $<$ or $>$: $-12 \square -21$

$>$

6 Order smallest to largest: $5, -8, 0, -3, -10$

$-10, -8, -3, 0, 5$

7 $-9 + 14$

5

8 $3 + (-11)$

-8

9 $-4 - 7$

-11

10 $-2 - (-10)$

8

11 -7×8

-56

12 $-6 \times (-9)$

54

13 $-81 \div 9$

-9

14 $-64 \div (-8)$

8

15 $-5 + 3 \times (-4)$

$= -5 + (-12) = -17$

16 $(-5 + 3) \times (-4)$

$= -2 \times (-4) = 8$

17 Calculator: $-384 \div 16$

-24

18 Calculator: $(-73 + 28) \times 11$

$= -45 \times 11 = -495$

19 It is -3°C . It falls 9°C . What is the temperature now?

$-3 - 9 = -12^\circ\text{C}$

20 A diver at -25 m rises 4 m a minute for 5 minutes. Where is she now?

$-25 + 5 \times 4 = -5$ m

21 How far apart are 40 m and -35 m?

$40 - (-35) = 75$ m

22 $-2 \times (-3) \times (-5)$

$= 6 \times (-5) = -30$

23 Is $6 - (-4)$ the same as $6 + 4$? Explain.

Yes. Subtracting -4 is adding its opposite, 4 . Both are 10 .

24 Put brackets in to make it true: $2 - 8 \times 3 = -18$

$(2 - 8) \times 3 = -18$